

## Genome-wide association resources

### Microbiome GWAS

18,340 participants  
211 taxonomic traits

### Species GWAS

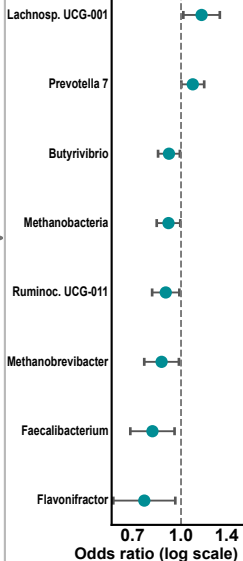
16,017 participants  
17 metagenomic traits

### Essential-tremor GWAS

16,480 cases  
1,936,173 controls

## Mendelian randomization

Eight distinct genus-level instrument sets



## Chemically resolved metabolites

|                         |                   |
|-------------------------|-------------------|
| Acid<br>98              | $\gamma$ VL<br>97 |
| (5R)- $\gamma$ VL<br>96 | 4-OH acid<br>97   |
| 3'-S<br>97              | 4'-S<br>97        |
| 3'-G<br>96              | 4'-G<br>96        |

targets per exact form

## SwissTargetPrediction targets

|                                                                 |
|-----------------------------------------------------------------|
| 198<br>Predicted in $\geq 1$ form                               |
| 150<br>Predicted for $\geq 1$ sulfate/<br>glucuronide conjugate |
| 45<br>Shared by all sulfate/<br>glucuronide conjugates          |
| 25<br>Predicted in<br>all 8 forms                               |

CA1 CA2 CA9 CA12: 8/8 forms  
CA3: 5/8 forms

## Essential-tremor genetic and cerebellar analyses

**CA3** Gene-based association  
Granule-cell cis-eQTL  
Oligodendrocyte cis-eQTL

**CA12** Higher expression in  
granule cells, Bergmann glia  
and astrocytes

**CA7** Purkinje-cell  
differential expression

109 donors | 1,004,112 nuclei